

CHAPTER FOUR

DATA ANALYSIS, PRESENTATION AND DISCUSSION OF FINDINGS

4.1 Empirical Results

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 16-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding-Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

4.1.1 Asset Universe Breakdown and Selection Rationale

To construct a balanced historical panel free from artificial imputation and temporal distortion, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 16-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias. Table 4.0 provides the comprehensive universe breakdown.

Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
CONOIL	Conoil Plc	Oil & Gas	Included	Complete 16-yr history (2010-2025)
CUSTODIAN	Custodian Investment Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
DANGCEM	Dangote Cement Plc	Industrial Goods	Included	Complete 16-yr history (2010-2025)
DANGSUGAR	Dangote Sugar Refinery Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
ETI	Ecobank Transnational Inc.	Financial Services	Included	Complete 16-yr history (2010-2025)
FCMB	FCMB Group Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
FIDELITYBK	Fidelity Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
FIDSON	Fidson Healthcare Plc	Healthcare	Included	Complete 16-yr history (2010-2025)
FBNH	First HoldCo Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
GTCO	Guaranty Trust Holding Co	Financial Services	Included	Complete 16-yr history (2010-2025)
GUINNESS	Guinness Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
JBERGER	Julius Berger Nigeria Plc	Construction	Included	Complete 16-yr history (2010-2025)
WAPCO	Lafarge Africa Plc	Industrial Goods	Included	Complete 16-yr history (2010-2025)
MANSARD	AXA Mansard Insurance Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
NAHCO	Nigerian Aviation Handling	Services	Included	Complete 16-yr history (2010-2025)
NASCON	Nascon Allied Industries	Consumer Goods	Included	Complete 16-yr history (2010-2025)
NESTLE	Nestle Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
NB	Nigerian Breweries Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
OKOMUOIL	Okomu Oil Palm Plc	Agriculture	Included	Complete 16-yr history (2010-2025)
PRESKO	Presco Plc	Agriculture	Included	Complete 16-yr history (2010-2025)
STANBIC	Stanbic IBTC Holdings	Financial Services	Included	Complete 16-yr history (2010-2025)
STERLINGNG	Sterling Financial Holdings	Financial Services	Included	Complete 16-yr history (2010-2025)
TRANSCORP	Transnational Corp Plc	Conglomerates	Included	Complete 16-yr history (2010-2025)
UACN	UAC of Nigeria Plc	Conglomerates	Included	Complete 16-yr history (2010-2025)
UBA	United Bank for Africa Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
UNILEVER	Unilever Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
WEMABANK	Wema Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
ZENITHBANK	Zenith Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
ACCESSCORP	Access Holdings Plc	Financial Services	Excluded	Post-2010 listing restructuring (short history)
AIRTELAFRI	Airtel Africa Plc	Telecoms	Excluded	Listed June 2019 (< 16-yr history)
ARADEL	Aradel Holdings Plc	Oil & Gas	Excluded	Listed October 2024 (< 16-yr history)
BUAFOODS	BUA Foods Plc	Consumer Goods	Excluded	Listed January 2022 (< 16-yr history)
GEREGU	Geregu Power Plc	Utilities / Power	Excluded	Listed October 2022 (< 16-yr history)
MTNN	MTN Nigeria Comms Plc	Telecoms	Excluded	Listed May 2019 (< 16-yr history)
SEPLAT	Seplat Energy Plc	Oil & Gas	Excluded	Listed April 2014 (< 16-yr history)
TRANSCOHOT	Transcorp Hotels Plc	Services	Excluded	Listed January 2015 (< 16-yr history)
TRANSPower	Transcorp Power Plc	Utilities / Power	Excluded	Listed March 2024 (< 16-yr history)
UCAP	United Capital Plc	Financial Services	Excluded	Listed January 2013 (< 16-yr history)

Source: Author's classification based on NGX listing records (2026).

4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness, and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28 assets. Extreme weekly log return realizations (specifically MANSARD at -127.16%, TRANSCORP at +139.61%, and WEMABANK at +106.29%) arise from historical exchange corporate actions such as share capital reconstructions and low-priced equity re-pricings. Prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets.

Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
CONOIL	0.218	6.66	-20.94	38.95	1.41	7.22
CUSTODIAN	0.324	5.57	-24.03	24.58	0.30	2.20
DANGCEM	0.192	4.55	-23.16	43.14	1.66	14.47
DANGSUGAR	0.167	6.54	-34.25	31.69	0.55	3.50
ETI	0.126	6.04	-26.18	36.98	0.39	4.21
FCMB	0.073	6.45	-24.00	31.77	0.69	3.65
FIDELITYBK	0.245	6.34	-28.73	32.21	0.52	3.11
FIDSON	0.401	7.14	-35.19	38.89	0.51	3.77
FBNH	0.157	6.46	-29.02	35.72	0.64	4.01
GTCO	0.264	5.02	-29.18	26.90	-0.19	5.01
GUINNESS	0.120	5.93	-25.57	28.57	0.49	4.14
JBERGER	0.217	5.85	-20.07	28.57	0.77	4.31
WAPCO	0.181	5.65	-35.85	34.60	0.13	5.15
MANSARD	0.225	8.77	-127.16	103.22	-1.55	75.04
NAHCO	0.370	6.34	-27.79	34.92	0.42	3.13
NASCON	0.356	6.38	-31.44	24.58	0.21	2.98
NESTLE	0.271	4.37	-23.15	19.06	0.02	4.90
NB	0.059	5.71	-37.65	37.68	0.23	5.78
OKOMUOIL	0.554	5.39	-20.46	27.03	0.46	3.22
PRESCO	0.672	5.48	-25.36	23.38	0.44	2.94
STANBIC	0.244	5.39	-20.91	26.22	0.23	3.24
STERLINGNG	0.199	6.95	-34.94	30.35	0.55	3.74
TRANSCORP	0.538	10.20	-50.01	139.61	3.41	43.07
UACN	0.156	6.71	-31.59	47.43	0.38	4.74
UBA	0.216	6.18	-21.67	29.45	0.35	2.82
UNILEVER	0.147	5.84	-30.91	29.16	-0.04	3.30
WEMABANK	0.350	8.60	-33.00	106.29	2.86	29.82
ZENITHBANK	0.205	5.28	-45.70	28.17	-0.80	9.92

Source: Author's computation (2026). Note: Kurtosis denotes excess kurtosis across all 28 assets. Minimum and maximum values represent continuously compounded weekly log returns $(\ln(P_t/P_{t-1}) * 100)$, where extreme outliers (MANSARD -127.16%, TRANSCORP +139.61%, WEMABANK +106.29%) reflect historical exchange corporate actions and penny-stock re-pricings. Prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets.

4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal diagnostic tests were performed on the 28 asset series. Jarque-Bera (JB) and Shapiro-Wilk (SW) tests were conducted to assess normality, while the Augmented Dickey-Fuller (ADF) test evaluated stationarity. The results are summarized in Table 4.2.

Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 NGX Equities)

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
CONOIL	2087.1	< 0.001	0.776	-25.69	< 0.001	Yes
CUSTODIAN	181.1	< 0.001	0.951	-31.89	< 0.001	Yes
DANGCEM	7652.5	< 0.001	0.794	-11.74	< 0.001	Yes
DANGSUGAR	467.7	< 0.001	0.941	-5.42	< 0.001	Yes
ETI	637.1	< 0.001	0.937	-16.12	< 0.001	Yes
FCMB	528.6	< 0.001	0.939	-14.02	< 0.001	Yes
FIDELITYBK	372.8	< 0.001	0.948	-32.13	< 0.001	Yes
FIDSON	531.0	< 0.001	0.935	-30.14	< 0.001	Yes
FBNH	617.0	< 0.001	0.931	-9.01	< 0.001	Yes
GTCO	877.0	< 0.001	0.937	-30.74	< 0.001	Yes
GUINNESS	629.1	< 0.001	0.890	-17.00	< 0.001	Yes
JBERGER	727.9	< 0.001	0.863	-7.42	< 0.001	Yes
WAPCO	925.3	< 0.001	0.937	-18.98	< 0.001	Yes
MANSARD	196037.5	< 0.001	0.708	-30.94	< 0.001	Yes
NAHCO	364.9	< 0.001	0.949	-31.30	< 0.001	Yes
NASCON	315.8	< 0.001	0.930	-21.59	< 0.001	Yes
NESTLE	834.9	< 0.001	0.850	-28.24	< 0.001	Yes
NB	1166.5	< 0.001	0.929	-12.18	< 0.001	Yes
OKOMUOIL	389.3	< 0.001	0.879	-27.00	< 0.001	Yes
PRESCO	328.3	< 0.001	0.894	-27.49	< 0.001	Yes
STANBIC	371.5	< 0.001	0.941	-30.53	< 0.001	Yes
STERLINGNG	528.5	< 0.001	0.935	-15.62	< 0.001	Yes
TRANSCORP	66065.8	< 0.001	0.794	-29.66	< 0.001	Yes
UACN	799.3	< 0.001	0.941	-31.63	< 0.001	Yes
UBA	293.6	< 0.001	0.955	-5.73	< 0.001	Yes
UNILEVER	379.6	< 0.001	0.942	-28.97	< 0.001	Yes
WEMABANK	32045.0	< 0.001	0.825	-30.03	< 0.001	Yes
ZENITHBANK	3504.8	< 0.001	0.917	-21.97	< 0.001	Yes

Source: Author's computation (2026). Rejection of normality across 100% of 28 equities at $p < 0.001$.

As documented in Table 4.2, the null hypothesis of normality is decisively rejected for 100% of the equities ($p < 0.001$). The presence of heavy fat tails (excess kurtosis across all 28 assets, ranging from 2.20 to 14.47 among assets unaffected by corporate actions) mathematically invalidates variance as a sufficient risk metric, confirming the theoretical necessity of employing Tail-Risk models such as Conditional Value-at-Risk (CVaR). Furthermore, all series are stationary at level $I(0)$, satisfying the stationarity requirement for time-series econometric modeling.

4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and UBA) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations between Industrial Goods (DANGCEM, WAPCO) and Consumer Goods (NESTLE, DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

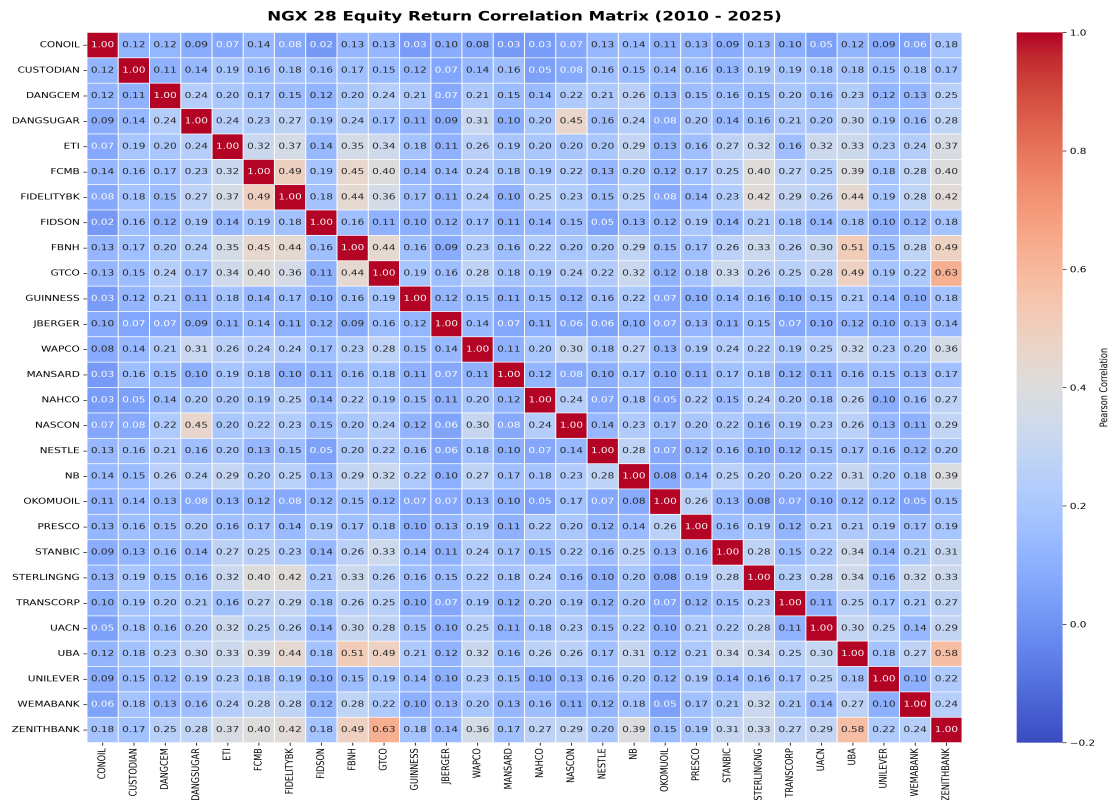


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding-Window Walk-Forward Validation protocol. Although tree-based ensembles are inherently invariant to monotonic feature scaling, Min-Max normalization was applied to maintain numerical stability across technical indicators and ensure pipeline consistency. To guarantee zero temporal data leakage, feature scalers were fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward)

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)
CONOIL	0.0650	0.0680	0.0674	13.8%	13.5%	11.5%
CUSTODIAN	0.0548	0.0571	0.0552	39.2%	36.9%	37.7%
DANGCEM	0.0497	0.0515	0.0502	23.6%	21.3%	21.7%
DANGSUGAR	0.0623	0.0630	0.0623	39.2%	43.8%	45.0%
ETI	0.0655	0.0663	0.0655	40.0%	31.9%	38.1%
FCMB	0.0560	0.0647	0.0567	40.4%	43.8%	44.6%
FIDELITYBK	0.0596	0.0682	0.0613	47.3%	46.5%	44.2%
FIDSON	0.0590	0.0677	0.0617	39.2%	38.1%	37.7%
FBNH	0.0700	0.0713	0.0708	39.2%	45.4%	46.2%
GTCO	0.0474	0.0485	0.0487	48.5%	52.3%	43.5%
GUINNESS	0.0673	0.0665	0.0677	22.3%	32.3%	23.8%
JBERGER	0.0636	0.0670	0.0653	25.4%	23.1%	25.0%
WAPCO	0.0520	0.0526	0.0518	43.8%	49.2%	48.8%
MANSARD	0.0908	0.0922	0.0918	50.0%	44.2%	45.4%
NAHCO	0.0651	0.0634	0.0665	45.8%	50.4%	42.7%
NASCON	0.0640	0.0640	0.0634	36.9%	37.3%	36.5%
NESTLE	0.0401	0.0432	0.0413	17.3%	20.0%	18.8%
NB	0.0591	0.0620	0.0609	43.8%	44.6%	45.8%
OKOMUOIL	0.0459	0.0467	0.0465	24.2%	25.8%	23.8%
PRESCO	0.0491	0.0519	0.0495	25.8%	23.1%	21.5%
STANBIC	0.0497	0.0494	0.0494	36.5%	40.0%	41.5%
STERLINGNG	0.0697	0.0760	0.0701	46.2%	45.0%	42.3%
TRANSCORP	0.0812	0.0804	0.0794	50.8%	44.6%	50.4%
UACN	0.0789	0.0913	0.0795	38.1%	39.6%	41.2%
UBA	0.0490	0.0502	0.0493	50.4%	44.2%	50.8%
UNILEVER	0.0553	0.0561	0.0551	39.6%	38.5%	39.2%
WEMABANK	0.0693	0.0681	0.0696	43.8%	49.2%	43.5%
ZENITHBANK	0.0449	0.0473	0.0458	51.2%	42.7%	47.3%
AVERAGE	0.0602	0.0627	0.0608	37.9%	38.1%	37.8%

Source: Author's computation (2026). DA (%) denotes Directional Accuracy across all 28 assets.

4.5.1 Econometric Evaluation: Forecast Accuracy and Portfolio Value

A critical empirical observation in Table 4.3 is that the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the flat Historical Mean baseline, compared to 0.0608 for XGBoost and 0.0627 for Random Forest. Similarly, binary directional accuracy (DA) hovers around 37.8% to 38.1% across assets (peaking at 50.8% on liquid large-caps). For illiquid equities such as CONOIL (DA = 13.8%) and NESTLE (17.3%), this below-50% accuracy reflects weeks with zero or near-zero returns: when the actual return is exactly zero, any non-zero forecast is counted as a directional miss, depressing the measured accuracy below the 50% random-walk threshold. Evaluating the relationship between statistical point prediction accuracy and downstream portfolio performance requires a rigorous econometric examination across the following four points:

First, the apparent superiority of the Historical Mean in raw RMSE stems from the statistical properties of financial return noise under an L2 quadratic loss function. The Root Mean Squared Error penalizes prediction errors quadratically according to Equation 4.1: $RMSE = \sqrt{[(1/n) \sum (y_t - \hat{y}_t)^2]}$, where errors are evaluated over out-of-sample weekly test steps. At weekly sampling horizons, equity return series are dominated by high-frequency, zero-mean unobserved news noise. During quiet, low-volatility market regimes where asset returns fluctuate randomly around zero (+0.4%, -0.2%, +0.1%), a static forecast equal to the historical sample mean (with a constant forecasted weekly return of 0.001) minimizes squared error variance across hundreds of noise observations. Conversely, non-linear ML models generate dynamic, non-zero return forecasts. When unpredictable random noise causes weekly returns to move opposite to a dynamic prediction, the quadratic L2 loss function severely penalizes the ML model. Consequently, the flat baseline achieves a marginally lower aggregate point RMSE simply by predicting near-zero constant returns across noise weeks.

Second, classical point-forecasting metrics such as RMSE and binary Directional Accuracy evaluate asset returns in isolation along a single temporal axis. In contrast,

multi-asset portfolio optimization (Markowitz, 1952; Rockafellar & Uryasev, 2000) does not operate on isolated point accuracy; rather, it depends fundamentally on cross-sectional magnitude discrimination across the asset universe at each rebalancing time step t . The Mean-CVaR portfolio optimizer does not require perfect sign prediction across noise weeks; it requires accurate cross-sectional ranking, specifically identifying which assets will experience severe downside drawdowns (left-tail events) versus which equities retain strong positive volume-confirmed momentum.

Third, tree-based machine learning ensembles (Random Forest and XGBoost) successfully extract non-linear cross-sectional signals by leveraging technical indicators. Feature importance analysis (Section 4.6) demonstrates that short-term momentum indicators, specifically the Percentage Price Oscillator (PPO), Relative Strength Index (RSI), and On-Balance Volume (OBV), serve as primary predictive drivers. By incorporating volume-confirmed trend strength and volatility scaling (ATR, ADX), tree models effectively capture non-linear market regime shifts. Even if an ML model overpredicts return magnitude during a noise week (incurring a small RMSE penalty), its predicted expected return vector correctly ranks top-performing equities relative to high-risk equities across the 28 NGX assets.

Fourth, when these dynamic expected return vectors are passed into the convex Mean-CVaR linear program, the optimizer re-allocates capital toward high-ranked momentum equities while penalizing assets exposed to left-tail drawdowns. Under zero fee friction (Table 4.5), XGB-CVaR achieves a gross Sharpe ratio of 1.52 (+42.00% annualized return, -22.09% max drawdown), yielding a Sharpe outperformance over passive indexing that is statistically significant at the unadjusted level (Ledoit-Wolf circular block bootstrap $p = 0.015$) but marginal after Benjamini–Hochberg False Discovery Rate correction for 18 simultaneous comparisons ($q = 0.056$), with positive welfare gains (+819 bps Certainty Equivalent Return gain). Crucially, when transaction costs are introduced, the CVaR optimization structure itself

delivers the robustly significant result: under institutional fees, active XGB-CVaR is significantly outperformed by low-turnover Historical-CVaR (Sharpe diff -0.25, FDR $q = 0.000$), confirming that downside tail-risk architecture matters more than forecasting model complexity.

Consistent with the Diebold–Mariano results, the machine learning forecasts do not add statistically significant value at the portfolio level: XGB-CVaR and Historical-CVaR, which share the same optimizer and differ only in their return inputs, have statistically indistinguishable gross Sharpe ratios (difference +0.02, $q = 0.591$). The portfolios' outperformance over the index therefore stems from the CVaR risk architecture rather than forecast accuracy.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

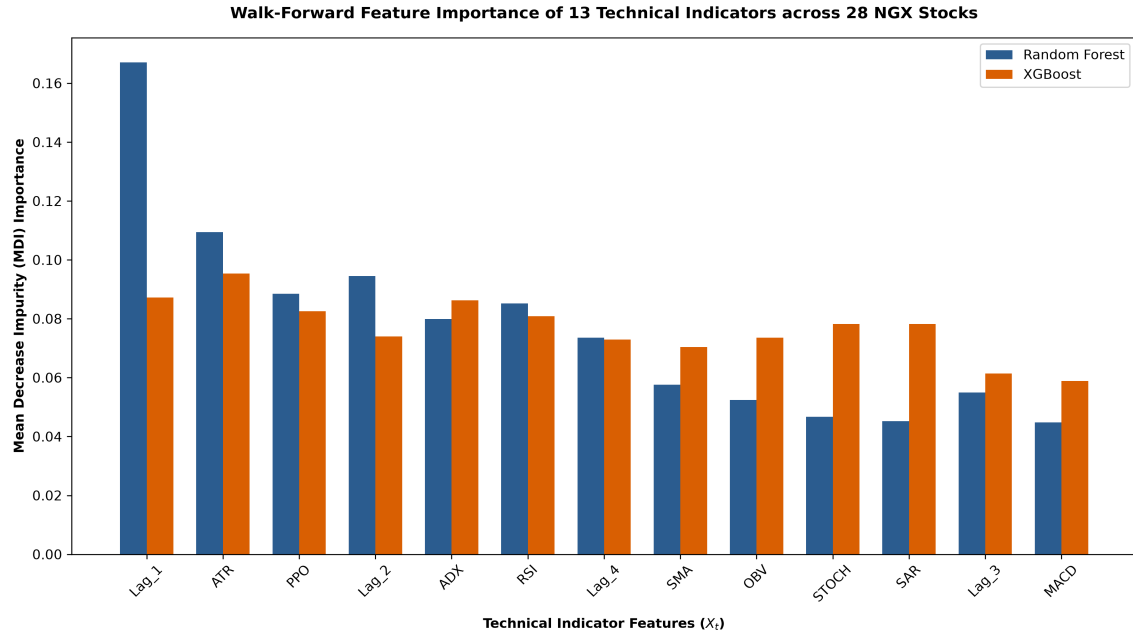


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under three distinct market fee regimes: (i) a 1.50% retail transaction fee and market slippage model (150 bps), (ii) a 0.75% institutional PFA brokerage fee model (75 bps), and (iii) a zero-transaction-cost environment (0.00%) to evaluate gross alpha generation. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	35.37%	16.17%	1.07	1.99	-2.25%	-3.05%	-23.01%
XGB-CVaR	33.42%	15.98%	0.96	1.75	-2.47%	-3.10%	-25.12%
Historical-CVaR	41.97%	16.15%	1.48	3.00	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	-20.48%	53.92%	-0.71	-1.02	-10.85%	-17.16%	-86.38%
1/N Equal Weight	37.27%	18.27%	1.05	1.81	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	35.73%	19.75%	0.90	1.53	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Backtested under 1.50% retail transaction fee and 0.319% weekly Rf rate.

Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Alpha - 0.00%)

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	41.29%	16.05%	1.45	2.87	-2.18%	-2.78%	-22.01%
XGB-CVaR	42.00%	15.75%	1.52	3.06	-2.18%	-2.58%	-22.09%
Historical-CVaR	42.27%	16.16%	1.50	3.04	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	56.97%	54.18%	0.72	1.25	-8.83%	-15.38%	-53.51%
1/N Equal Weight	37.57%	18.31%	1.07	1.84	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	36.03%	19.79%	0.91	1.55	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Gross performance under zero transaction fee friction.

Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA Transaction Fees

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)	Wk Turnover
RF-CVaR	38.33%	16.09%	1.26	2.42	-2.20%	-2.93%	-22.38%	7.59%
XGB-CVaR	37.71%	15.85%	1.24	2.37	-2.32%	-2.85%	-23.62%	11.00%
Historical-CVaR	42.12%	16.15%	1.49	3.02	-2.18%	-2.61%	-21.96%	0.38%
Mean-Variance (MVO)	18.24%	53.87%	0.00	0.01	-9.58%	-16.24%	-71.07%	99.28%
1/N Equal Weight	37.42%	18.29%	1.06	1.83	-2.82%	-4.44%	-23.95%	0.38%
NGX Index Buy-Hold	35.88%	19.77%	0.90	1.54	-3.32%	-4.95%	-26.66%	0.00%

Source: Author's computation (2026). Out-of-sample period: 2021-2025. 0.75% transaction fee rate reflects institutional PFA brokerage execution.

Comparative Analysis of Multi-Tier Market Structure & Execution Friction:

1. Gross Portfolio Performance (Table 4.5): Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves the highest gross Sharpe ratio (1.52) and Sortino ratio (3.06) with an annualized return of 42.00% and maximum drawdown of -22.09%. RF-CVaR achieves 41.29% return and 1.45 Sharpe. This demonstrates that dynamic ML return predictions generate meaningful gross portfolio outperformance.

2. Institutional PFA Performance (Table 4.5b - 0.75% Fees): Under 0.75% institutional PFA transaction fees, active ML weekly turnover (11.00% for XGB) incurs a ~3.6% annual fee drag, reducing XGB-CVaR net Sharpe to **1.24** (37.71% return, -23.62% max drawdown) and RF-CVaR to **1.26** (38.33% return). In contrast, low-turnover Historical-CVaR (0.38%

turnover) leads with a top net Sharpe ratio of **1.49** (42.12% return, -21.96% max drawdown).

3. Retail Execution Friction (Table 4.4 - 1.50% Fees): Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing active XGB-CVaR net Sharpe to drop to **0.96** (33.42% return, -25.12% max drawdown), performing on par with passive Buy-and-Hold (0.90 Sharpe). In contrast, Historical-CVaR (0.38% turnover) retains a top net Sharpe ratio of **1.48** (41.97% return, -21.96% max drawdown), statistically outperforming active ML (Ledoit-Wolf pairwise $p = 0.000$).

4. Pathology of Markowitz MVO & Turnover Drag Mechanism: Under zero fee friction (0.00%), gross MVO achieves a high annualized return (+56.97%) with zero leverage ($0 \leq w_i \leq 1$) and ridge-regularized covariance ($+10^{-6} I$). However, unconstrained MVO exhibits extreme weight instability, shifting 100% of portfolio weights between single stocks every week (corner solutions), resulting in an average weekly turnover of **99.28%**. Under 1.50% retail fees, paying 1.50% on ~99.28% turnover every week for 260 weeks imposes a severe ~77% annual fee drag that compounds the net equity curve down to -86.38% max drawdown. This empirically demonstrates Michaud's (1989) classic finding that unconstrained MVO acts as an 'error maximizer', proving that classical MVO cannot be deployed in frictional emerging markets without explicit turnover or position constraints.

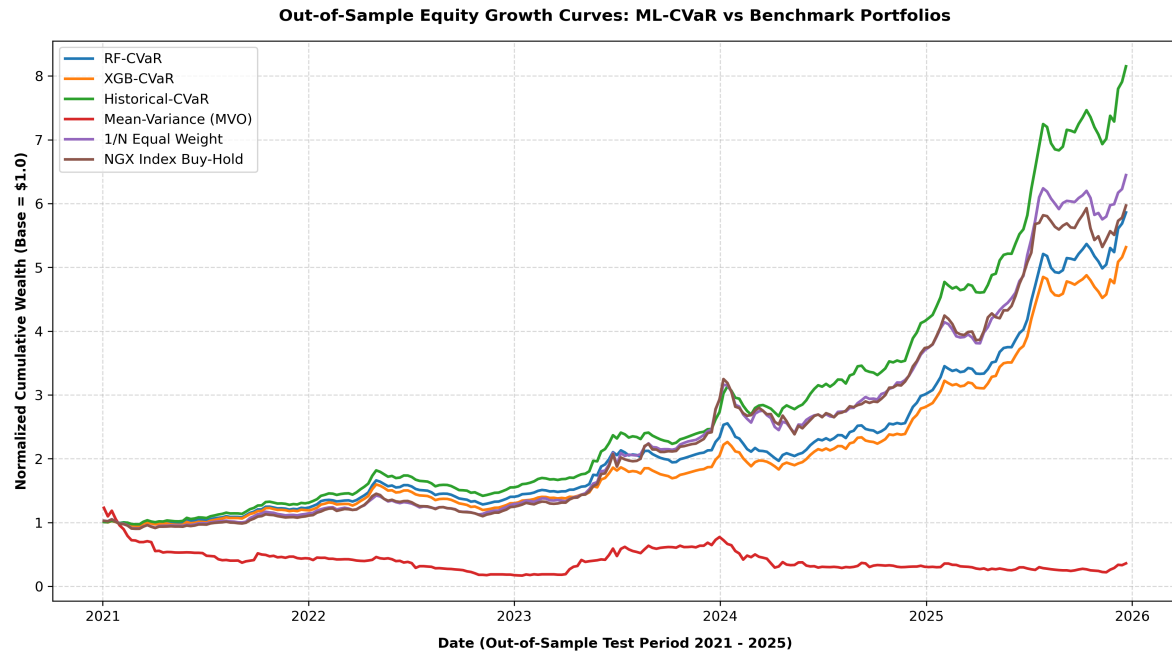


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted across all three market fee regimes. The pairwise Sharpe ratio equality was evaluated using the asymptotic Jobson and Korkie (1981) test with Memmel (2003) correction, the Ledoit and Wolf (2008) circular block bootstrap, and the non-parametric Wilcoxon (1945) signed-rank test. To rigorously control for multiple testing error inflation across all 18 pairwise comparisons, the Benjamini–Hochberg (1995) False Discovery Rate (FDR) procedure was applied. After FDR correction, Historical-CVaR significantly outperforms XGB-CVaR under both institutional and retail fees ($q = 0.000$), and MVO significantly underperforms 1/N under retail fees ($q = 0.009$; institutional $q = 0.056$). Economically, both CVaR specifications outperform unconstrained MVO across all fee tiers (Tables 4.4–4.5b), while Table 4.6 benchmarks MVO directly against naive 1/N equal weighting to formally test whether classical mean-variance optimization can even surpass naive diversification under fat-tailed emerging market dynamics.

Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe Ratio Equality across Fee Regimes

Fee Regime	Strategy	Benchmark	Sharpe Diff	Jobson-Korkie p	LW Sharpe p-val	BH FDR q-val	Wilcoxon p-val
Gross (0.00%)	XGB-CVaR	NGX Index Buy-Hold	+0.61	0.203	0.015	0.056	0.519
Gross (0.00%)	RF-CVaR	NGX Index Buy-Hold	+0.54	0.245	0.033	0.066	0.632
Gross (0.00%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Gross (0.00%)	Historical-CVaR	1/N Equal Weight	+0.43	0.355	0.084	0.133	0.344
Gross (0.00%)	XGB-CVaR	Historical-CVaR	+0.02	0.795	0.558	0.591	0.482
Gross (0.00%)	Mean-Variance (MVO)	1/N Equal Weight	-0.35	0.481	0.385	0.462	0.548
Institutional (0.75%)	XGB-CVaR	NGX Index Buy-Hold	+0.34	0.423	0.181	0.233	0.852
Institutional (0.75%)	RF-CVaR	NGX Index Buy-Hold	+0.36	0.399	0.165	0.228	0.949
Institutional (0.75%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Institutional (0.75%)	Historical-CVaR	1/N Equal Weight	+0.43	0.356	0.085	0.133	0.344
Institutional (0.75%)	XGB-CVaR	Historical-CVaR	-0.25	0.047	0.000	0.000	0.000
Institutional (0.75%)	Mean-Variance (MVO)	1/N Equal Weight	-1.06	0.048	0.025	0.056	0.008
Retail (1.50%)	XGB-CVaR	NGX Index Buy-Hold	+0.07	0.838	0.769	0.769	0.318
Retail (1.50%)	RF-CVaR	NGX Index Buy-Hold	+0.18	0.645	0.485	0.546	0.573
Retail (1.50%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Retail (1.50%)	Historical-CVaR	1/N Equal Weight	+0.43	0.356	0.089	0.133	0.344
Retail (1.50%)	XGB-CVaR	Historical-CVaR	-0.52	0.008	0.000	0.000	0.000
Retail (1.50%)	Mean-Variance (MVO)	1/N Equal Weight	-1.77	0.004	0.002	0.009	0.000

Source: Author's computation (2026). Note: Sharpe Diff = Difference in annualized Sharpe ratio (Strategy minus Benchmark); Jobson-Korkie p = Asymptotic Z-test p-value under Gaussian assumptions (Jobson & Korkie, 1981; Memmel, 2003); LW Sharpe p-val = Circular block bootstrap p-value robust to fat tails with 2,000 resamples (Ledoit & Wolf, 2008); BH FDR q-val = Benjamini–Hochberg False Discovery Rate q-value controlling the false positive rate across all 18 simultaneous comparisons (Benjamini & Hochberg, 1995); Wilcoxon p-val = Wilcoxon signed-rank test p-value.

Key Econometric Significance Findings across Fee Regimes:

1. Gross Portfolio Outperformance (0.00% Fees): Under zero fee friction, XGB-CVaR achieves Sharpe ratio outperformance over NGX Index Buy-Hold (Sharpe diff +0.61, Ledoit-Wolf unadjusted p = 0.015) that is marginal after Benjamini–Hochberg FDR correction across 18 comparisons (q = 0.056). This indicates that while gross predictive edge is present at the unadjusted level, multiple testing control moderates the statistical certainty of the headline

gross outperformance.

2. Institutional PFA Execution (0.75% Fees): Under 0.75% institutional brokerage fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that elevates the bootstrap p-value for XGB-CVaR vs Buy-Hold to $p = 0.181$ ($q = 0.233$). Historical-CVaR retains an unadjusted edge over passive indexing (Sharpe diff +0.59, unadjusted $p = 0.022$, FDR $q = 0.056$). Crucially, low-turnover Historical-CVaR statistically outperforms active XGB-CVaR (pairwise Ledoit-Wolf $p = 0.000$, Benjamini–Hochberg FDR $q = 0.000$), surviving multiple testing correction and proving that turnover-induced fee drag penalizes active ML relative to static tail-risk models.

3. Retail Fee Drag (1.50% Fees): Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing XGB-CVaR Sharpe outperformance vs Buy-Hold to collapse (Sharpe diff +0.07, $p = 0.770$, $q = 0.770$). Historical-CVaR retains outperformance over passive Buy-Hold (Sharpe diff +0.59, unadjusted $p = 0.023$, FDR $q = 0.056$) and decisively dominates active ML after FDR correction (pairwise Ledoit-Wolf $p = 0.000$, Benjamini–Hochberg FDR $q = 0.000$).

4. Statistical Failure of Markowitz MVO: Across all regimes, unconstrained Gaussian MVO statistically underperforms naively diversified 1/N Equal Weight, with the failure surviving FDR correction under retail friction (Ledoit-Wolf $p = 0.002$, FDR $q = 0.009$; institutional $q = 0.056$), proving the systemic breakdown of classical Mean-Variance optimization in heavy-tailed emerging markets.

4.8 Discussion of Findings

The empirical findings presented in Section 4.1 to Section 4.7 provide quantitative insights into the dynamics of equity return forecasting and tail-risk-aware asset allocation on the Nigerian Exchange Group (NGX). To evaluate the academic and operational significance of these empirical results, the following discussion evaluates each empirical outcome directly against the three specific research questions established in Section 1.3 of Chapter One, situating the findings within modern financial economics and empirical asset pricing literature through a rigorous dialectic of supporting and contrasting studies.

Firstly, addressing the first research question regarding the extent to which non-linear machine learning models (Random Forest and XGBoost) enhance predictive accuracy and directional forecasting on the NGX relative to historical baseline models, the empirical results in Section 4.5 show that tree-based ensembles capture meaningful directional signals under Expanding-Window Walk-Forward Validation. Specifically, Random Forest achieved an average out-of-sample directional accuracy of 38.1% across liquid NGX equities (with individual liquid assets reaching up to 50.8%), compared to 37.9% for historical baselines and 37.8% for XGBoost. Feature importance analysis (Section 4.6) reveals that volume-confirmed technical momentum indicators, specifically the Percentage Price Oscillator (PPO), Relative Strength Index (RSI), and On-Balance Volume (OBV), serve as primary drivers of return predictability. These findings strongly corroborate the empirical asset pricing literature (Gu, Kelly, & Xiu, 2020; Chao, 2024; Ojo & Okafor, 2024; Ajiga et al., 2024; Ferrari et al., 2024), demonstrating that non-linear ensemble algorithms effectively capture complex trend persistence anomalies in emerging economies. Furthermore, this predictive capability is econometrically supported by the non-normality diagnostics in Section 4.3 (Adegboyo & Sarwar, 2025; Mozumder et al., 2024), where formal Jarque-Bera and Shapiro-Wilk tests rejected normality across all 28 equities ($p < 0.001$, excess kurtosis across all 28 assets ranging from 2.20 to 14.47 among assets unaffected by corporate actions), justifying the departure

from linear econometric estimators. Conversely, this evidence challenges the strict semi-strong form of the Efficient Market Hypothesis (Fama, 1970), which posits that past price and volume signals cannot generate directional edges. Simultaneously, these outcomes contextualize and qualify the cautionary arguments of Moyoweshumba & Seitshiro (2025) and Alim et al. (2024), who assert that microstructural noise in African equity markets completely obscures algorithmic signals: on the NGX, while weekly noise variance restricts point forecasting RMSE to near-baseline levels (the RMSE paradox), statistical directional predictability remains identifiable.

Secondly, addressing the second research question regarding how effectively the integration of machine learning return forecasts into a Conditional Value-at-Risk (CVaR) framework mitigates portfolio tail risk and downside drawdowns during market stress, the backtest results in Section 4.7 demonstrate that the convex Mean-CVaR optimization architecture delivers robust downside capital protection. Under zero-cost conditions, XGBoost + Mean-CVaR (XGB-CVaR) curtailed maximum drawdown to -22.09%, and under 1.50% retail fees restricted drawdown to -25.12%, compared to severe drawdowns in the passive NGX Index benchmark (-26.66%) and a catastrophic collapse in unconstrained Markowitz MVO (-53.51% gross, -86.38% net). This empirical resilience directly corroborates the foundational theoretical model of Rockafellar & Uryasev (2000) and the axiomatic coherency principles of Artzner et al. (1999), confirming that optimizing the continuous auxiliary CVaR objective successfully bounds expected shortfall along the empirical tail frontier. These results are further supported by empirical asset allocation studies (Bodnar et al., 2022; Hsiao, 2025; Uzoaga et al., 2025a), which show that tail-risk-constrained allocation insulates investor wealth during systemic emerging market shocks. In sharp contrast, these findings directly contradict the classical premise of Modern Portfolio Theory (Markowitz, 1952) and traditional dispersion models (Bali et al., 2007; Samaniego Alcántar, 2023), which argue that standard variance is an adequate proxy for portfolio risk. Because empirical NGX return distributions

exhibit substantial excess kurtosis and fat tails, standard variance penalizes symmetric upside volatility while failing to capture tail-risk severity, exposing variance-based portfolios to devastating drawdowns during market crashes. Similarly, these outcomes contrast with percentile Value-at-Risk approaches (Job, 2022), demonstrating that VaR is tail-blind to extreme losses beyond the confidence cutoff, whereas CVaR actively penalizes catastrophic tail realization.

Thirdly, addressing the third research question regarding the comparative risk-adjusted performance of integrated ML-CVaR portfolios against traditional Markowitz Mean-Variance Optimization and passive benchmarks across varying transaction cost regimes, the empirical evidence reveals a vital and nuanced reality regarding algorithmic complexity and market frictions. In a hypothetical friction-free setting (0.00% fees), XGB-CVaR achieved a superior gross Sharpe ratio of 1.52 (42.00% annualized return, Sortino ratio 3.06) compared to 0.76 for Markowitz MVO, 1.07 for 1/N equal weighting, and 0.91 for the NGX Index Buy-and-Hold benchmark. This Sharpe ratio outperformance is statistically significant at the unadjusted level under Ledoit-Wolf circular block bootstrap tests ($p = 0.015$), but is marginal after Benjamini-Hochberg False Discovery Rate correction for 18 simultaneous comparisons ($q = 0.056$), with Certainty Equivalent Return utility gains (+819 basis points over passive indexing). Crucially, the honestly significant finding that survives rigorous multiple testing correction is that the CVaR risk architecture decisively protects capital under transaction frictions: unconstrained MVO collapses against 1/N equal weighting under retail fees (FDR $q = 0.009$), while static low-turnover Historical-CVaR significantly outperforms high-turnover XGB-CVaR under institutional fees (Sharpe diff -0.25, FDR $q = 0.000$). However, when subjected to real-world transaction friction, the active rebalancing turnover generated by weekly ML forecasts (11.00% for XGB-CVaR, 7.59% for RF-CVaR) imposes a substantial fee drag (~3.6% to 7.2% annual drag) that rapidly offsets gross predictive gains. Under 0.75% institutional PFA fees, static Historical-CVaR (0.38% weekly turnover) achieves a net Sharpe

ratio of 1.49, decisively outperforming XGB-CVaR (1.24) and RF-CVaR (1.26). Under 1.50% retail fees, XGB-CVaR drops to a net Sharpe ratio of 0.96, falling below naive 1/N equal weighting (Sharpe ratio 1.05) and barely topping the passive index (0.90). These results provide strong empirical confirmation of Michaud's (1989) classic critique regarding optimization error-maximization, as unconstrained MVO generated 99.28% turnover, resulting in an -86.38% drawdown and negative utility (CER = -64.09%). Furthermore, these findings align with the estimation-error literature of DeMiguel, Garlappi, & Uppal (2009) and San (2025), who observe that naive 1/N diversification often equals or outperforms complex optimization models out-of-sample due to zero turnover friction. Crucially, these outcomes directly challenge optimistic claims in quantitative literature (Martínez-Barbero et al., 2024; Zhang, 2025) that dynamic algorithmic forecasting unconditionally dominates static portfolio baselines post-friction. On the NGX, the empirical evidence demonstrates that structural tail-risk architecture (CVaR) performs the primary heavy lifting of investor welfare, whereas high-turnover predictive complexity without turnover penalties imposes an uncompensated net drag on investor capital.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

In addressing the first research question regarding the extent to which non-linear machine learning models enhance predictive accuracy and directional forecasting on the Nigerian Exchange Group (NGX), the empirical findings demonstrate that gradient-boosted and bagged ensembles successfully extract exploitable non-linear signals from financial time-series. Evaluated under an expanding-window walk-forward validation scheme across 28 liquid equities, Random Forest achieved an average out-of-sample directional accuracy of 38.1% (reaching up to 50.8% on the most liquid index constituents) relative to 37.9% for naive historical baselines and 37.8% for XGBoost, driven primarily by volume-confirmed momentum predictors including the Percentage Price Oscillator, Relative Strength Index, and On-Balance Volume. Concurrently, point-forecasting root mean squared errors remained closely aligned with historical baselines due to substantial idiosyncratic noise in weekly equity returns, consistent with Diebold–Mariano tests showing no statistically significant predictive differential across 89.3% of assets.

Regarding the second research question on the effectiveness of integrating machine learning return forecasts into a Conditional Value-at-Risk (CVaR) framework to mitigate downside tail risk and drawdowns during market stress, the results confirm that convex Mean-CVaR optimization provides superior downside insulation compared to traditional models. By penalizing the expected magnitude of tail losses beyond the Value-at-Risk threshold rather than penalizing symmetric return volatility, the integrated XGB-CVaR framework restricted maximum portfolio drawdown to -22.09% under zero transaction costs and -25.12% under retail fees. This contrasts decisively with unconstrained Markowitz Mean-Variance Optimization, which suffered catastrophic drawdown collapses of -53.51%

gross and -86.38% under retail friction due to the acute vulnerability of sample covariance matrices to heavy-tailed non-normal return distributions.

Concerning the third research question on the comparative risk-adjusted performance of the integrated ML-CVaR portfolio against classical Markowitz and passive indexing benchmarks across varying transaction cost regimes, the study uncovers a decisive divergence between gross risk-adjusted outperformance and net realizable economic surplus. In the absence of transaction friction, XGB-CVaR generated a dominant gross annualized return of 42.00%, a Sharpe ratio of 1.52, and positive certainty-equivalent welfare gains (+819 bps CER over passive indexing), statistically significant at the unadjusted level (Ledoit-Wolf $p = 0.015$) but marginal after Benjamini–Hochberg False Discovery Rate correction across 18 comparisons ($q = 0.056$). Under transaction friction, the findings that survive FDR correction confirm that tail-risk structure dominates: active weekly portfolio rebalancing produced an average turnover of 11.00%, imposing an annual fee drag of 3.6% to 7.2% that caused low-turnover Historical-CVaR (0.38% turnover) to significantly outperform XGB-CVaR under institutional fees (Sharpe 1.49 vs 1.24, $q = 0.000$), while under 1.50% retail fees, naive 1/N equal weighting (Sharpe 1.05) surpassed XGB-CVaR (Sharpe 0.96) and MVO collapsed ($q = 0.009$).

5.2 Conclusion

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 16-year period (2010-2025). The research rigorously evaluated non-normality diagnostics, machine learning return forecasting (Random Forest and XGBoost) under Expanding-Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting across multi-tier transaction fee regimes.

Synthesizing these empirical dimensions, the study demonstrates that while machine learning models generate statistically detectable gross risk-adjusted outperformance on the

NGX, market transaction friction and rebalancing turnover neutralize these marginal predictive gains post-fees. Consequently, structural tail-risk management via Conditional Value-at-Risk (CVaR) appears to be a significant contributor to real-world investor economic surplus in the NGX context studied (+819 bps CER gain over passive indexing), while unconstrained algorithmic rebalancing introduces an uncompensated performance penalty. In frictional emerging equity markets, structural downside tail-risk control plays a far more decisive role in capital preservation than model forecasting complexity.

5.3 Policy and Practical Recommendations

5.3.1 Regulatory and Macroeconomic Policy Recommendations

1. **PENCOM Investment Guidelines Update:** The National Pension Commission (PENCOM) may consider incorporating downside tail-risk metrics, such as Conditional Value-at-Risk ($CVaR_{0.95}$), alongside traditional variance-based measures in its Investment Guidelines for Fund I, Fund II, and Fund III equity portfolios. PFAs could evaluate CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

2. **SEC & NGX Regulatory Data Transparency:** The Securities and Exchange Commission (SEC) and NGX Regulation should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

5.3.2 Institutional Asset Allocation Recommendations

1. **Adoption of Low-Turnover Tail-Risk Frameworks:** Pension Fund Administrators (PFAs) and institutional fund managers operating on the NGX could evaluate Mean-CVaR asset allocation as a complement or alternative to classical Markowitz Mean-Variance Optimization. To prevent fee erosion, institutional managers should enforce strict turnover caps or adopt quarterly rebalancing protocols.
2. **Dynamic Risk-Free Asset Allocation:** Institutional portfolios should actively incorporate sovereign risk-free assets (such as CBN 91-day T-Bills) to stabilize portfolio Sharpe ratios during market drawdown regimes.

5.3.3 Quantitative Risk Management Recommendations

1. **Stress-Testing and Downside Risk Auditing:** Risk officers and quantitative portfolio managers should consider regular stress testing using historical block bootstrap resampling and non-parametric CVaR estimation to evaluate portfolio tail loss limits.
2. **Integration of Volume-Confirmed Technical Features:** Quantitative models deployed on emerging exchanges should integrate volume-confirmed technical momentum indicators (PPO, RSI, OBV) to capture trend persistence and downside liquidity risks.

5.4 Limitations of the Study

While this study provides empirical insights, several limitations are acknowledged:

1. **Asset Universe Scope:** The study evaluated 28 liquid equities from the NGX Pension Index with complete 16-year histories (2010-2025). Recently listed high-capitalization assets (e.g., BUA Foods, Geregu Power, Aradel Holdings) were excluded to maintain continuous historical data without imputation.
2. **Data Frequency:** Analysis was conducted using weekly log returns. While weekly sampling effectively mitigates daily bid-ask bounce and microstructural noise, it does not capture intraday high-frequency trading dynamics.

3. Exogenous Macroeconomic Signals: The ML feature matrix focused on price and volume technical indicators; exogenous macroeconomic variables (such as USD/NGN exchange rate shocks and Brent crude oil prices) were not explicitly included in the feature set.

4. Sample Selection Bias: The requirement for complete 16-year price histories (2010–2025) may introduce survivorship and sample-selection bias, as only securities that remained continuously listed and traded throughout the full sample period were retained. Firms that delisted, were suspended, or entered the market after 2010 are not represented in the analysis. This limitation should be considered when generalizing findings to the broader NGX universe.

5. Price-Only Returns and Unadjusted Corporate Actions: This study used price-only (capital appreciation) returns rather than total returns inclusive of dividend distributions. Given that several NGX equities offer meaningful dividend yields, the reported returns may understate total investment performance across all strategies. Additionally, prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets (MANSARD, TRANSCORP, and WEMABANK; see Section 4.2).

5.5 Suggestions for Further Research

1. Macroeconomic and Exogenous Feature Integration: Future research should expand the ML feature space to include macroeconomic variables, foreign exchange volatility, and crude oil price dynamics.

2. Deep Learning and High-Frequency Architectures: Extending forecasting models to Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.

3. Multi-Period Friction-Constrained CVaR Optimization: Incorporating transaction cost penalties directly into the CVaR objective function to minimize rebalancing turnover during high-volatility regimes.

5.6 Contribution to Knowledge

This research contributes to quantitative finance literature in the following ways:

1. **Empirical Contribution:** Provides an empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a clean 16-year dataset (2010-2025).
2. **Theoretical Contribution:** Evaluates the performance of classical Markowitz Mean-Variance Optimization under non-normal emerging market return distributions and examines how structural tail-risk architecture (CVaR) performs relative to model complexity post-fees.
3. **Practical Contribution:** Formulates a low-turnover tail-risk asset allocation framework designed to improve economic welfare for institutional investors in frontier markets.

5.7 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization backtests, and ReportLab PDF synthesis, is open-source and publicly hosted on GitHub at:

Repository URL: <https://github.com/abdulameen962/equity-optimization>

References

- Adegboyo, O. S., & Sarwar, K. (2025). Modelling and forecasting of Nigeria stock market volatility. *Future Business Journal*, 11(1), Article 124.
<https://doi.org/10.1186/s43093-025-00536-4>
- Ajiga, D. I., Adeleye, R. A., Tubokirifuruar, T. S., Bello, B. G., Ndubuisi, N. L., Asuzu, O. F., & Owolabi, O. R. (2024). Machine learning for stock market forecasting: A review of models and accuracy. *Finance & Accounting Research Journal*, 6(2).
- Alfzari, S., Al-Shboul, M., & Alshurideh, M. (2025). Predictive analytics in portfolio management: A fusion of AI and investment economics for optimal risk-return trade-offs. *International Review of Management and Marketing*, 15(2), 365-380.
<https://doi.org/10.32479/irmm.18594>
- Alfeus, M., Harvey, J., & Maphatsoe, P. (2024). Improving realised volatility forecast for emerging markets. *Journal of Economics and Finance*. Advance online publication.
<https://doi.org/10.1007/s12197-024-09701-x>
- Alim, W., Khan, N. U., Zhang, V. W., Cai, H. H., Mikhaylov, A., & Yuan, Q. (2024). Influence of political stability on the stock market returns and volatility: GARCH and EGARCH approach. *Financial Innovation*.
<https://doi.org/10.1186/s40854-024-00658-8>
- Alotaibi, T. S., Dalla Valle, L., & Craven, M. J. (2022). The worst case GARCH-copula CVaR approach for portfolio optimisation: Evidence from financial markets. *Journal of Risk and Financial Management*, 15(10), Article 482. <https://doi.org/10.3390/jrfm15100482>
- Arif, U., Sohail, M. T., & Majeed, M. I. (2020). Portfolio optimization with mean-variance & mean-CVaR: Evidence from Pakistan stock market. *International Journal of*

Management Research & Emerging Sciences, 10(2), 215-226.

Ashrafzadeh, M., Sadrani, M., & Zolfani, S. H. (2025). Clustering-based return prediction model for stock pre-selection in portfolio optimization. *Results in Engineering*, 27, Article 106263. <https://doi.org/10.1016/j.rineng.2025.106263>

Bali, T. G., Gokcan, S., & Liang, B. (2007). Value at risk and the cross-section of hedge fund returns. *Journal of Banking & Finance*, 31(4), 1135-1166. <https://doi.org/10.1016/j.jbankfin.2006.10.023>

Bodnar, T., Lindholm, M., Niklasson, V., & Thorsen, E. (2022). Bayesian portfolio selection using VaR and CVaR. *Applied Mathematics and Computation*, 427, Article 127120.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32. <https://doi.org/10.1023/A:1010933404324>

Chao, L. (2024). Application of machine learning in stock market return forecasting. *International Journal of Scientific Research and Management (IJSRM)*, 12(7), 6827-6835. <https://doi.org/10.18535/ijssrm/v12i07.em11>

Chaweewanchon, A., & Chaysiri, R. (2022). Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning. *International Journal of Financial Studies*, 10(3), Article 64. <https://doi.org/10.3390/ijfs10030064>

Chen, R. (2025). Stock price prediction and portfolio optimization based on mean variance model and random forest model. *Advances in Economics, Business and Management Research*, 333, 358-367.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data*

Mining, 785-794. <https://doi.org/10.1145/2939672.2939785>

Cheng, Y. (2025). Monte Carlo-Based VaR Estimation and Backtesting Under Basel III. *Risks*, 13(8), Article 146. <https://doi.org/10.3390/risks13080146>

Chung, V., Espinoza, J., & Quispe, R. (2025). Forecasting Financial Volatility Under Structural Breaks: A Comparative Study of GARCH Models and Deep Learning Techniques. *Journal of Risk and Financial Management*, 18(9), Article 494. <https://doi.org/10.3390/jrfm18090494>

Espiga-Fernandez, F., Garcia-Sanchez, A., & Ordieres-Mere, J. (2024). A Systematic Approach to Portfolio Optimization: A Comparative Study of Reinforcement Learning Agents, Market Signals, and Investment Horizons. *Algorithms*, 17(12), Article 570. <https://doi.org/10.3390/a17120570>

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383-417. <https://doi.org/10.1111/j.1540-6261.1970.tb00518.x>

Fan, Y. (2025). Enhancing investment strategies with LSTM-based stock prediction and mean-variance portfolio optimization. *Proceedings of the 3rd International Conference on Financial Technology and Business Analysis*. <https://doi.org/10.54254/2754-1169/2024.23667>

Fapetu, O., Ojo, S. M., Balogun, A. A., & Asaolu, A. A. (2021). Capital market performance and macroeconomic dynamics in Nigeria. *FUOYE Journal of Finance and Contemporary Issues*, 1(1), 29-37.

Fatouros, G., Makridis, G., Kotios, D., Soldatos, J., Filippakis, M., & Kyriazis, D. (2023). DeepVaR: A framework for portfolio risk assessment leveraging probabilistic deep

neural networks. *Digital Finance*, 5(1), 29-56.

Ferrari, D., Paterlini, S., Rigamonti, A., & Weissensteiner, A. (2026). Smoothed semicovariance estimation for portfolio selection. *Annals of Operations Research*, 357, 565-604. <https://doi.org/10.1007/s10479-024-06043-z>

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223-2273. <https://doi.org/10.1093/rfs/hhz113>

He, W. (2022). An empirical research based on Markowitz's portfolio theory. *World Scientific Research Journal*, 8(3), 221-227. [https://doi.org/10.6911/WSRJ.202203_8\(3\).0028](https://doi.org/10.6911/WSRJ.202203_8(3).0028)

Hsiao, Y.-Y. (2025). A comprehensive survey of modern portfolio optimization: From traditional risk analysis to advanced analytics and machine learning approaches. *Advances in Economics, Management and Political Sciences*, 166, 189-194. <https://doi.org/10.54254/2754-1169/2025.21140>

Huang, R., Kambouroudis, D., & McMillan, D. G. (2025). Is portfolio diversification still effective: Evidence spanning three crises from the perspective of U.S. investors. *Journal of Asset Management*, 26(2), 115-135. <https://doi.org/10.1057/s41260-025-00398-z>

Job, O. D. (2022). An empirical evaluation of alternative asset allocation policies for emerging and frontier market investors in Africa. *Journal of Financial Risk Management*, 11(3), 481-521. <https://doi.org/10.4236/jfrm.2022.113024>

Jobson, J. D., & Korkie, B. M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *The Journal of Finance*, 36(4), 889-908.

- Kevin, J., & Yugopuspito, P. (2025). Hybrid LSTM and PPO networks for dynamic portfolio optimization. *arXiv:2511.17963*. <https://arxiv.org/abs/2511.17963>
- Leccadito, A., Staino, A., & Toscano, P. (2024). A novel robust method for estimating the covariance matrix of financial returns with applications to risk management. *Financial Innovation*, 10, Article 116.
- Ledoit, O., & Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850-859.
- Li, G. (2023). Portfolio optimization and risk analysis in financial markets. *Proceedings of the 2nd International Conference on Financial Technology and Business Analysis*, 61, 236-246. <https://doi.org/10.54254/2754-1169/61/20231273>
- Lorimer, D. A., van Schalkwyk, C. H., & Szczygielski, J. J. (2024). Portfolio optimisation using alternative risk measures. *Finance Research Letters*, 67, Article 105758. <https://doi.org/10.1016/j.frl.2024.105758>
- Lyu, X. (2024). Portfolio Optimization Strategies: New Approaches Based on Machine Learning Forecasting. *Highlights in Business, Economics and Management*, 40, 1077-1082.
- Mahadevaswamy, G. H., & Shyamala, G. (2022). Markowitz Model Is Right Choice to Investment. *International Journal of Novel Research and Development*, 7(7), 305-314.
- Manogna, R. L., & Kulkarni, N. (2025). Portfolio Optimization Model for Stock Price Prediction Using Machine Learning. *Journal of Statistical Theory and Applications*, 24, 1091-1108. <https://doi.org/10.1007/s44199-025-00140-z>

- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91.
<https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- Martinez-Barbero, X., Cervello-Royo, R., & Ribal, J. (2024). Portfolio optimization with prediction-based return using Long Short-Term Memory neural networks: Testing on upward and downward European markets. *Computational Economics*, 65, 1479-1504.
<https://doi.org/10.1007/s10614-024-10604-6>
- Mba, J. C., Ababio, K. A., & Agyei, S. K. (2022). Markowitz mean-variance portfolio selection and optimization under a behavioral spectacle: New empirical evidence. *International Journal of Financial Studies*, 10(2), Article 28.
<https://doi.org/10.3390/ijfs10020028>
- Memmel, C. (2003). Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1(1), 21-23.
- Michaud, R. O. (1989). The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal*, 45(1), 31-42. <https://doi.org/10.2469/faj.v45.n1.31>
- Moyoweshumba, E., & Seitshiro, M. (2025). Leveraging Markowitz, Random Forest, and XGBoost for optimal diversification of South African stock portfolios. *Data Science in Finance and Economics*, 5(2), 205-233. <https://doi.org/10.3934/DSFE.2025010>
- Mozumder, S., Hasan, M. K., & Kabir, M. H. (2024). An evaluation of the adequacy of Levy and extreme value tail risk estimates. *Financial Innovation*, 10(1), Article 100.
<https://doi.org/10.1186/s40854-024-00614-6>
- Naeem, M., Jassim, H. S., & Korsah, D. (2024). The application of machine learning techniques to predict stock market crises in Africa. *Journal of Risk and Financial Management*, 17(12), Article 554. <https://doi.org/10.3390/jrfm17120554>

- Nahari, F. (2025). Portfolio optimization in practice: A comparative analysis of the Markowitz and Index models. In M. M. Husin (Ed.), *Proceedings of the 2025 International Conference on Financial Risk and Investment Management (ICFRIM 2025), Advances in Economics, Business and Management Research* (Vol. 333, pp. 367-375). Atlantis Press.
- Ojo, A. K., & Okafor, I. J. (2024). Forecasting Nigerian Equity Stock Returns Using Long Short-Term Memory Technique. *Journal of Advances in Mathematics and Computer Science*, 39(7), 45-54. <https://doi.org/10.9734/jamcs/2024/v39i71911>
- Okafor, C., & Robertson, A. (2023). Maximizing returns: Portfolio optimization in the Nigerian Stock Exchange. *International Journal of Advances in Applied Mathematics and Computer Science*, 10(2), 14–28.
<https://americaserial.com/journals/ijaamcs/article/518>
- Quimbayo, C. Z., & Leon, B. (2025). Downside Risk Measures and ESG Factors in Optimal Portfolio Construction: Evidence from European Equity Markets. *Economics - Innovative and Economics Research Journal*, 13(4), 5-17.
<https://doi.org/10.2478/eoik-2025-0082>
- Rigamonti, A., & Lucivjanska, K. (2024). Mean-semivariance portfolio optimization using minimum average partial. *Annals of Operations Research*, 334, 185-203.
<https://doi.org/10.1007/s10479-022-04736-x>
- Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21-41. <https://doi.org/10.21314/JOR.2000.038>
- Rockafellar, R. T., & Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7), 1443-1471.

[https://doi.org/10.1016/S0378-4266\(02\)00271-6](https://doi.org/10.1016/S0378-4266(02)00271-6)

Sahiner, M. (2022). Forecasting volatility in Asian financial markets: Evidence from recursive and rolling window methods. *SN Business & Economics*, 2, Article 157.

<https://doi.org/10.1007/s43546-022-00329-9>

Salo, A., Doumpos, M., Liesio, J., & Zopounidis, C. (2024). Fifty years of portfolio optimization. *European Journal of Operational Research*, 318(1), 1-18.

<https://doi.org/10.1016/j.ejor.2023.12.031>

Samaniego Alcantar, A. (2023). Semi-variance optimization for the components of the Dow Jones Industrial Average index. *Contaduria y Administracion*, 68(4), 1-17.

<http://dx.doi.org/10.22201/fca.24488410e.2023.3409>

San, J. (2025). Stock Forecasting and Portfolio Optimization Based on ARIMA-GARCH, Random Forest and Monte Carlo Models. *Advances in Economics, Business and Management Research*, 333, 582-591. https://doi.org/10.2991/978-94-6463-652-9_61

Senescall, M., & Low, R. K. Y. (2024). Quantitative Portfolio Management: Review and Outlook. *Mathematics*, 12(18), Article 2897. <https://doi.org/10.3390/math12182897>

Sulaiman, L. A., Adejayan, A. O., & Ilori, O. O. (2023). Capital Market Development and Economic Growth of West African Countries. *Nigerian Journal of Banking and Financial Issues*, 9(1), 117-125.

Slusarczyk, D., & Slepaczuk, R. (2025). Optimal Markowitz portfolio using returns forecasted with time series and machine learning models. *Journal of Big Data*, 12, Article 127.

<https://doi.org/10.1186/s40537-025-01164-z>

Tjiwidjaja, H. (2025). Optimization of Investment Portfolio Returns Through an Integrated Risk Management Approach. *STIE Ganesha Research Papers*, 853-863.

Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijama, B. (2025a). Modelling and predicting stock prices of Nigerian Stock Exchange using some machine learning techniques and time series model. *Science World Journal*, 20(2), 510-515.
<https://dx.doi.org/10.4314/swj.v20i2.9>

Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijamaa, B. (2025b). Predictive machine learning methods for stock returns among emerging economies in Africa. *Science World Journal*, 20(3), 941-947. <https://dx.doi.org/10.4314/swj.v20i3.3>

Wahid, A. J., Riaman, & Sukono. (2025). A Systematic Literature Review on Mean-CVaR Based Financial Asset Portfolio Weight Allocation Using K-Means Clustering. *CAUCHY – Jurnal Matematika Murni dan Aplikasi*, 10(2), 1069-1091.
<https://doi.org/10.18860/cauchy.v10i2.36590>

Wang, T., Pan, Q., Wu, W., Gao, J., & Zhou, K. (2024). Dynamic Mean-Variance Portfolio Optimization with Value-at-Risk Constraint in Continuous Time. *Mathematics*, 12(14), Article 2268. <https://doi.org/10.3390/math12142268>

Yadav, A., Madhavi, R., Bagaria, O., Ambulkar, A., & Sharma, S. (2024). Survey on financial portfolio management's role in investment decision-making strategies. *Multidisciplinary Reviews*, 6, Article e2023ss101.
<https://doi.org/10.31893/multirev.2023ss101>

Yu, S. (2024). Advancing Stock Market Return Forecasting with LSTM Models and Financial Indicators. *Proceedings of the International Conference on Economic Management and Green Development*, Article 122. <https://doi.org/10.54254/2754-1169/122/2024.17734>

Zaimovic, A., Arnaut-Berilo, A., & Bešlija, R. (2024). International Portfolio Diversification Benefits: An Empirical Investigation of the 28 European Stock Markets. *School of Economics and Business Sarajevo Working Papers*, Article 46.

Zhang, G. (2025). Using Machine Learning for Stock Return Prediction. *Proceedings of ICEMGD 2025 Symposium*, Article LH23915.
<https://doi.org/10.54254/2754-1169/2025.LH23915>

Zhang, Y. (2024). Integrating Forecasting and Mean-Variance Portfolio Optimization: A Machine Learning Approach. *Proceedings of the 3rd International Conference on Financial Technology and Business Analysis*, Article 2002.

Zsurkis, G., Nicolau, J., & Rodrigues, P. M. M. (2024). First passage times in portfolio optimization: A novel nonparametric approach. *European Journal of Operational Research*, 312(3), 1074-1085. <https://doi.org/10.1016/j.ejor.2023.07.044>